1)

I want you for president.

w d d d d d d

i w

I wanna u 4 present.

w : wrong
i : inserted
d : deleted

⇒ Levenshtein distance :

$$\begin{array}{r} 2 \cdot w \\ 1 \cdot i \\ 6 \cdot d \\ \hline 9 \end{array}$$

2) • in the case of letter based distance:

additional blanks are additional insertions

and are additional errors

• in the case of word error based distance:

blanks are ignored

no change in distance.

3)

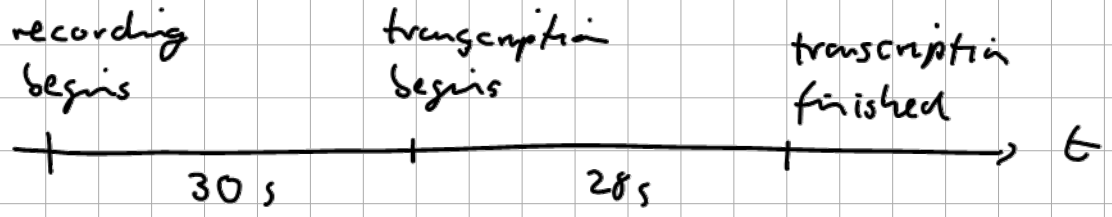
	r	e	f	e	r	e	n	c	e		w	o	r	d
r														
e														
c														
o														
r														
d														
e														
d														
w														
o														
r														
d														

distance matrix

$\xrightarrow{\text{Deletions}}$

$\downarrow$ Insertions

4)



Latency = 30 s + 28 s = 58 s

real time factor = $\frac{28}{30} < 1$

algorithm is capable of real time analysis.

5) a) No, the whisper takes too long. Real time for human users is somewhere in the region of 100 ms up to a maximum of 200 ms of latency.

6) Sampling rate for whisper is fixed at 16 kHz.

7) Vector x has a certain range of magnitudes.

By the given normalization vector y gets the same range of magnitudes as input vector x.

8) a) Variations of words must be recorded.

By this, the number of words in the dataset increases a lot.

Variations are e.g. conjugates like
- eine
- einer
- eines
- ...

b) The prosody of a spoken sentence differs from single words.

Examples:
- in a question, the last word is usually spoken in a higher pitch.
- The short breaks between words recorded and on its own makes the voice sounding machine like without any emotions.

9) $$p(n) = \frac{1}{n \ln(1.78 \cdot N)}$$

„der" $$p(1) = \frac{1}{\ln(1.78 \cdot 10^5)} = 6{,}95\,\%$$

„die" $$p(2) = \frac{1}{2 \ln(1.78 \cdot 10^5)} = 3{,}47\,\%$$

„und" $$p(3) = \frac{1}{3 \ln(1.78 \cdot 10^5)} = 2{,}32\,\%$$

„in" $$p(4) = \frac{1}{4 \ln(1.78 \cdot 10^5)} = 1{,}74\,\%$$

„den" $$p(5) = \frac{1}{5 \cdot \ln(1.78 \cdot 10^5)} = 1{,}39\,\%$$

$$\sum_{n=1}^{n_i} \frac{1}{n \ln(1.78 \cdot 10^5)} = 0{,}5$$

$$\sum_{n=1}^{n_i} \frac{1}{n} = \ln(1.78 \cdot 10^5) \cdot 0{,}5 = 7{,}1961$$

solution in python: 238 words

$$n_i = 238$$

10)

s_1 = K i s t e n

s_2 = K a r s t

Initialisation of distance matrix:

	#	K	a	r	s	t
#	0	1	2	3	4	5
K	1					
i	2					
s	3					
t	4					
e	5					
n	6					

	#	K	a	r	s	t
#	0	1	2	3	4	5
K	1	0	1	2	3	4
i	2	1	1	2	3	4
s	3	2	2	2	2	3
t	4	3	3	3	3	2
e	5	4	4	4	4	3
n	6	5	5	5	5	4

Levenshtein distance = 4